Classical Field Theory — Exercise Sheet 28

Tensor variation, gravitational waves and boundary action

Solutions

Use Lecture 28's curvature, inverse-metric and action conventions. All metrics, variations and coordinate maps are smooth. Repeated spacetime indices run from zero to three. Write $\eta = \mathrm{diag}(1,-1,-1,-1)$ only for the flat background. None of the problems asserts global existence for Einstein evolution.

Exercise 28.1 (The full Palatini tensor calculation). Let g_s be a smooth family of Lorentz metrics in fixed coordinates and put

$$k^{\mu\nu} = \partial_s g_s^{\mu\nu}\big|_0, \qquad h_{\mu\nu} = \partial_s (g_s)_{\mu\nu}\big|_0 = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}, \qquad C^{\rho}_{\mu\nu} = \delta\Gamma^{\rho}_{\mu\nu}.$$

Here ∇ is the background Levi–Civita connection.

(a) Prove that C is a tensor of type $(1,2)$ and derive its expression in terms of ∇h. Do not assume that connection coefficients themselves are tensor components.

(b) Vary the full coordinate Riemann formula. Expand both covariant derivatives of C, including all its index terms, to prove $\delta R^{\rho}{}_{\sigma\mu\nu} = \nabla_\mu C^{\rho}_{\nu\sigma} - \nabla_\nu C^{\rho}_{\mu\sigma}$. Contract to obtain the Palatini identity without setting $\Gamma = 0$.

(c) Contract again to derive the inverse-metric boundary vector $v^\mu = \nabla^\mu(g_{\alpha\beta}k^{\alpha\beta}) - \nabla_\nu k^{\mu\nu}$. This is the complete tensor argument promised by Lecture 28.

Solution. For two connections, their difference evaluated on X, Y is linear over smooth functions in X. It is also linear over such functions in Y: the two Leibniz terms $X(f)Y$ cancel when replacing Y by fY. Thus $\nabla^{(s)} - \nabla$ is a $(1,2)$ tensor. Its parameter derivative at zero, C, is a tensor as well. Torsion-freeness gives $C^{\rho}_{\mu\nu} = C^{\rho}_{\nu\mu}$.

Differentiating metric compatibility gives

$$\nabla_\mu h_{\nu\lambda} = C^{\rho}_{\mu\nu}g_{\rho\lambda} + C^{\rho}_{\mu\lambda}g_{\nu\rho}.$$

Add its (μ,λ,ν) and (ν,λ,μ) forms and subtract its (λ,μ,ν) form. The lower-index symmetry of C leaves

$$C^{\rho}_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\nabla_\mu h_{\lambda\nu} + \nabla_\nu h_{\lambda\mu} - \nabla_\lambda h_{\mu\nu}).$$

The full coordinate variation of Riemann is

$$\delta R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu C^{\rho}_{\nu\sigma} - \partial_\nu C^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}C^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}C^{\lambda}_{\mu\sigma} + C^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - C^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}.$$

On the other hand, the tensor covariant derivative is

$$\nabla_\mu C^{\rho}_{\nu\sigma} = \partial_\mu C^{\rho}_{\nu\sigma} + \Gamma^{\rho}_{\mu\lambda}C^{\lambda}_{\nu\sigma} - \Gamma^{\lambda}_{\mu\nu}C^{\rho}_{\lambda\sigma} - \Gamma^{\lambda}_{\mu\sigma}C^{\rho}_{\nu\lambda}.$$

Subtract its version with μ,ν interchanged. The terms with $\Gamma^{\lambda}_{\mu\nu}$ cancel by torsion-freeness. The remaining lower-σ terms and upper-index terms are precisely the four connection-product terms in the coordinate variation. This proves $\delta R^{\rho}{}_{\sigma\mu\nu} = \nabla_\mu C^{\rho}_{\nu\sigma} - \nabla_\nu C^{\rho}_{\mu\sigma}$ in arbitrary coordinates.

Using $\mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$ now gives

$$\delta \,\mathrm{Ric}_{\mu\nu} = \nabla_\rho C^\rho_{\nu\mu} - \nabla_\nu C^\rho_{\rho\mu}.$$

Metric compatibility allows the next contraction to be written as a divergence:

$$g^{\mu\nu}\delta \,\mathrm{Ric}_{\mu\nu} = \nabla_\rho\big(g^{\mu\nu}C^\rho_{\mu\nu} - g^{\rho\mu}C^\nu_{\mu\nu}\big).$$

Let $\tau = g_{\mu\nu}k^{\mu\nu}$. Substitution of $h_{\mu\nu} = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}$ into the formula for C yields

$$g^{\alpha\beta}C^\mu_{\alpha\beta} = -\nabla_\nu k^{\mu\nu} + \frac{1}{2}\nabla^\mu\tau, \qquad g^{\mu\alpha}C^\beta_{\alpha\beta} = -\frac{1}{2}\nabla^\mu\tau.$$

Their difference is $v^\mu = \nabla^\mu\tau - \nabla_\nu k^{\mu\nu}$, proving the claimed boundary-vector formula and its divergence identity.

Exercise 28.2 (Vacuum waves, gauge and the two polarisations). Set $g_{\mu\nu} = \eta_{\mu\nu} + sh_{\mu\nu} + O(s^2)$. Raise perturbation indices with η, write $h = \eta^{\mu\nu}h_{\mu\nu}$ and define $\bar{h}_{\mu\nu} = h_{\mu\nu} - \eta_{\mu\nu}h/2$.

(a) Derive $\Gamma^{(1)}$, $\mathrm{Ric}^{(1)}$ and $G^{(1)}$. Show explicitly that $G^{(1)}_{00}$ and $G^{(1)}_{0i}$ contain no second time derivatives of the perturbation. These are the linearized vacuum constraint equations on a slice.

(b) Under the covariant-metric Lie variation $h_{\mu\nu} \mapsto h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, derive the trace-reversed change and its divergence. In harmonic gauge $\partial^\mu\bar{h}_{\mu\nu} = 0$, derive the vacuum wave equation. No analytic theorem establishing that gauge globally is assumed or requested.

(c) For a mode $h_{\mu\nu} = \epsilon_{\mu\nu}e^{-\mathrm{i}\nu t + \mathrm{i}\nu x^3}$ with $\nu > 0$, impose harmonic gauge. Use residual null-wave gauge parameters to set the four temporal amplitudes to zero and derive the complete transverse traceless matrix. Count the independent polarisations. The real field is obtained by taking real parts; the plane mode itself is not claimed to have finite total energy on $\mathbb{R}^3$.

Solution. The first-order connection is

$$(\Gamma^{(1)})^\rho_{\mu\nu} = \frac{1}{2}\eta^{\rho\lambda}(\partial_\mu h_{\lambda\nu} + \partial_\nu h_{\lambda\mu} - \partial_\lambda h_{\mu\nu}).$$

The background connection is zero, so differentiating this formula gives

$$\mathrm{Ric}^{(1)}_{\mu\nu} = \frac{1}{2}(\partial_\rho\partial_\mu h^\rho{}_\nu + \partial_\rho\partial_\nu h^\rho{}_\mu - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h).$$

Its trace is $R^{(1)} = \partial_\rho\partial_\sigma h^{\rho\sigma} - \Box h$, and $G^{(1)} = \mathrm{Ric}^{(1)} - \eta R^{(1)}/2$. Expanding the time-index components, with spatial repeated indices summed, gives

$$\begin{aligned} G^{(1)}_{00} &= \frac{1}{2}(\Delta h_{jj} - \partial_i\partial_j h_{ij}), \\ G^{(1)}_{0i} &= \frac{1}{2}\left[\Delta h_{0i} - \partial_i\partial_j h_{0j} + \partial_t(\partial_i h_{jj} - \partial_j h_{ji})\right]. \end{aligned}$$

All second time derivatives have cancelled. The vacuum equations setting these expressions to zero constrain slice fields and their first time derivatives, rather than determining second time derivatives.

Trace reversal rewrites the full linearized tensor as

$$G^{(1)}_{\mu\nu} = \frac{1}{2}\left[\partial_\mu\partial_\rho\bar{h}^\rho{}_\nu + \partial_\nu\partial_\rho\bar{h}^\rho{}_\mu - \Box\bar{h}_{\mu\nu} - \eta_{\mu\nu}\partial_\rho\partial_\sigma\bar{h}^{\rho\sigma}\right].$$

The Lie variation has trace $2\partial_\rho\xi^\rho$ and hence

$$\delta\bar{h}_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu - \eta_{\mu\nu}\partial_\rho\xi^\rho, \qquad \partial^\mu\delta\bar{h}_{\mu\nu} = \Box\xi_\nu.$$

Thus harmonic gauge reduces vacuum to $\Box\bar{h}_{\mu\nu} = 0$. Parameters solving $\Box\xi_\nu = 0$ preserve that gauge. This algebra does not assert global solvability of an arbitrary gauge-fixing problem. Pure gauge perturbations give zero linearized curvature because their connection variation is $\partial_\mu\partial_\nu\xi^\rho$, whose antisymmetrised derivative is zero.

For the specified mode, harmonic gauge is $\ell^\mu\bar{\epsilon}_{\mu\nu} = 0$, where $\ell^\mu = (\nu, 0, 0, \nu)$ and $\ell_\mu\ell^\mu = 0$. A parameter with the same null phase changes the covariant amplitude by

$$\delta\epsilon_{\mu\nu} = -\mathrm{i}(\ell_\mu\alpha_\nu + \ell_\nu\alpha_\mu).$$

It preserves harmonic gauge. Set

$$\alpha_0 = \frac{\epsilon_{00}}{2\mathrm{i}\nu}, \quad \alpha_1 = \frac{\epsilon_{01}}{\mathrm{i}\nu}, \quad \alpha_2 = \frac{\epsilon_{02}}{\mathrm{i}\nu}, \quad \alpha_3 = \alpha_0 + \frac{\epsilon_{03}}{\mathrm{i}\nu}.$$

Since $\nu > 0$, these are defined and make every $\epsilon'_{0\mu} = 0$. Let $s_{\mathrm{sp}} = \epsilon'_{11} + \epsilon'_{22} + \epsilon'_{33}$. The remaining harmonic equations are $\bar{\epsilon}'_{0\nu} + \bar{\epsilon}'_{3\nu} = 0$. Their $\nu = 0$ component gives $s_{\mathrm{sp}} = 0$. Their $\nu = 1, 2$ components give $\epsilon'_{31} = \epsilon'_{32} = 0$. The $\nu = 3$ component then gives $\epsilon'_{33} = 0$. Thus $\epsilon'_{22} = -\epsilon'_{11}$ and the full amplitude is

$$\epsilon'_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The two free positive-frequency amplitudes encode two polarisations. For a real solution their negative-frequency partners are conjugate, rather than additional independent polarisations.

Exercise 28.3 (The non-null boundary action). Let Σ be a smooth non-null boundary patch of a region D. All variations have compact support in a collar of the interior of this patch and vanish near its edges and every other boundary piece. A Gaussian collar for the background may be used as a geometric given:

$$g = \epsilon_n\,\mathrm{d}n^2 + \gamma_{ab}(n, y)\mathrm{d}y^a\mathrm{d}y^b, \qquad \Sigma = \{n = 0\}, \quad D \text{ locally has } n \le 0, \quad \epsilon_n = \pm 1.$$

The outward unit normal is $N = \partial_n$. Write $\mathrm{vol}_\Sigma = \iota_N\mathrm{vol}_g$ with outward-first orientation, $K = \nabla_\mu N^\mu$, and fix only the induced metric: $h_{ab}|_\Sigma = 0$ for $h_{\mu\nu} = \delta g_{\mu\nu}$. Normal components of h are initially unrestricted.

Gaussian normal coordinates for a smooth non-null hypersurface are the local construction described in Wald, *General Relativity*, §3.3, pp. 42–43. The boundary-action discussion is Appendix E.1, pp. 458–459. Use the explicit conventions in this question, including ϵ_n.

(a) Construct a vector field ξ vanishing on Σ such that $\widetilde{h} = h + \mathcal{L}_\xi g$ has $\widetilde{h}_{nn} = \widetilde{h}_{na} = 0$ near the patch. Derive and solve the required first-order normal equations. Explain why the induced variation is still zero.

(b) In this representative compute the normal flux of the Palatini vector and the variation of $2\epsilon_n\int_\Sigma K\mathrm{vol}_\Sigma$. Prove their cancellation.

(c) Use diffeomorphism invariance and Lecture 28's Bianchi identity to extend that result to the original arbitrary normal components. State the resulting gravitational action with the frozen overall sign. Specify why this is not a null-boundary or corner/joint formula.

Solution. For the background collar the normal Lie components are

$$(\mathcal{L}_\xi g)_{nn} = 2\epsilon_n\partial_n\xi^n, \qquad (\mathcal{L}_\xi g)_{na} = \gamma_{ab}\partial_n\xi^b + \epsilon_n\partial_a\xi^n.$$

Define ξ by the normal initial-value equations

$$\partial_n \xi^n = -\frac{\epsilon_n}{2} h_{nn}, \qquad \xi^n(0,y) = 0,$$
$$\partial_n \xi^b = -\gamma^{ba}(h_{na} + \epsilon_n \partial_a \xi^n), \quad \xi^b(0,y) = 0.$$

First integrate the first line from zero to n. Then substitute that smooth solution and integrate the second line. The resulting ξ is smooth and has the desired zero boundary value. A cutoff away from the boundary extends it into D without changing these equations near the patch. Tangential compact support is preserved. On Σ, both ξ and its tangential derivatives vanish, so $(\mathcal{L}_\xi g)_{ab} = 0$ there. Thus $\widetilde{h_{ab}}|_\Sigma = 0$ as well.

Now use the Gaussian-preserving representative and write it simply as h. The inverse variation is $k^{\mu\nu} = -h^{\mu\nu}$, with the indices of h raised by the background. The first problem gives the equivalent covariant-variation vector

$$v^\mu = \nabla_\nu h^{\mu\nu} - \nabla^\mu (g^{\alpha\beta} h_{\alpha\beta}).$$

The normal components $h_{n\mu}$ vanish throughout a collar, and all components vanish on the boundary because the induced variation is fixed. Hence connection-times-h terms vanish at the boundary, and

$$v^n = -\epsilon_n \gamma^{ab} \partial_n h_{ab}.$$

The boundary density is $\epsilon_n (N_\mu v^\mu) \mathrm{vol}_\Sigma = v^n \mathrm{vol}_\Sigma$, because $N_\mu = \epsilon_n (\mathrm{d}n)_\mu$. This flux formula follows here directly from $\nabla_\mu v^\mu \mathrm{vol}_g = \partial_\mu (\sqrt{|\det g|} v^\mu) \mathrm{d}^4x$: normal-coordinate FTC leaves $v^n \mathrm{vol}_\Sigma$ at $n = 0$, while tangential integrals vanish by compact support.

In the same collar, $K = \frac{1}{2} \gamma^{ab} \partial_n \gamma_{ab}$. At the boundary $\delta\gamma^{ab} = 0$ and $\delta \mathrm{vol}_\Sigma = 0$. The normal vector remains ∂_n for this representative. Therefore

$$\delta K = \frac{1}{2} \gamma^{ab} \partial_n h_{ab}, \qquad \delta \left(2\epsilon_n \int_\Sigma K \mathrm{vol}_\Sigma \right) = \epsilon_n \int_\Sigma \gamma^{ab} \partial_n h_{ab} \mathrm{vol}_\Sigma.$$

This cancels the full normal-derivative contribution from the curvature integral, for either value of ϵ_n.

It remains to justify using this representative. The local flow of ξ fixes the boundary pointwise. The curvature integral and the oriented normal/extrinsic-curvature boundary integral are geometric integrals over the same region and patch; change of variables leaves each invariant under this boundary-fixing flow. Thus adding $\mathcal{L}_\xi g$ changes neither first variation of their sum. The bulk Einstein pairing is also unchanged. Its Lie contribution is

$$-2 \int_D G_{\mu\nu} \nabla^\mu \xi^\nu \, \mathrm{vol}_g = 0 :$$

integration by parts uses $\nabla^\mu G_{\mu\nu} = 0$ and a boundary current containing $\xi = 0$. Consequently the cancellation proves, for the original fixed-induced-metric variation too,

$$\delta \left[\int_D R_{\mathrm{sc}} \mathrm{vol}_g + 2\epsilon_n \int_\Sigma K \mathrm{vol}_\Sigma \right] = \int_D G_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g.$$

Multiplying by $-(16\pi G_\mathrm{N})^{-1}$ gives the stated gravitational action with boundary term $-(8\pi G_\mathrm{N})^{-1} \epsilon_n \int_\Sigma K \mathrm{vol}_\Sigma$. For compactly supported metric tests, adding Lecture 28's Hilbert matter variation gives the same interior equation $G_{\mu\nu} = 8\pi G_\mathrm{N} T_{\mu\nu}$.

The argument needs a smooth non-null patch and an invertible induced metric for its normal coordinates and inverse γ^{ab}. The support restriction removes edge and other-boundary contributions. It proves no formula for null boundaries or nonvanishing variations at corners/joints; their geometric data and boundary terms require separate analysis.